ASSIGNMENT-11.2

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BATCH-29

TASK-1:

PROMPT:

"""Stack data

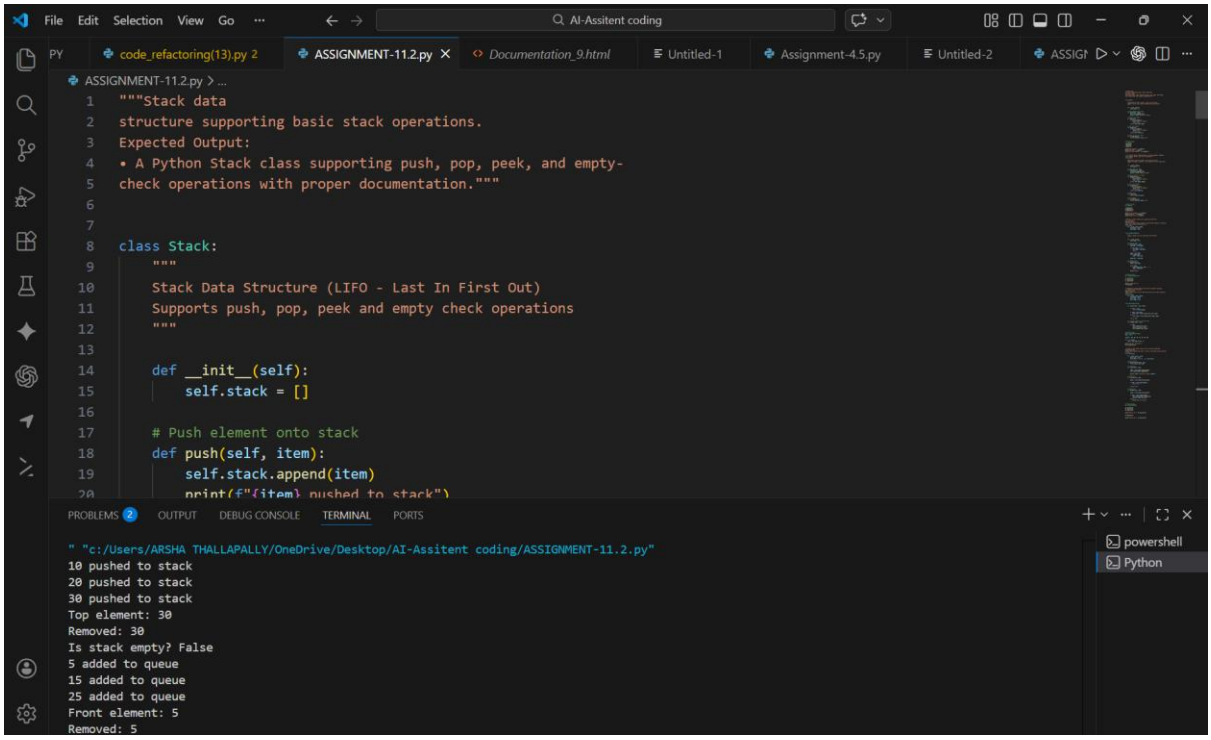
structure supporting basic stack operations.

Expected Output:

• A Python Stack class supporting push, pop, peek, and empty-

check operations with proper documentation."""

CODE:



OBSERVATION:

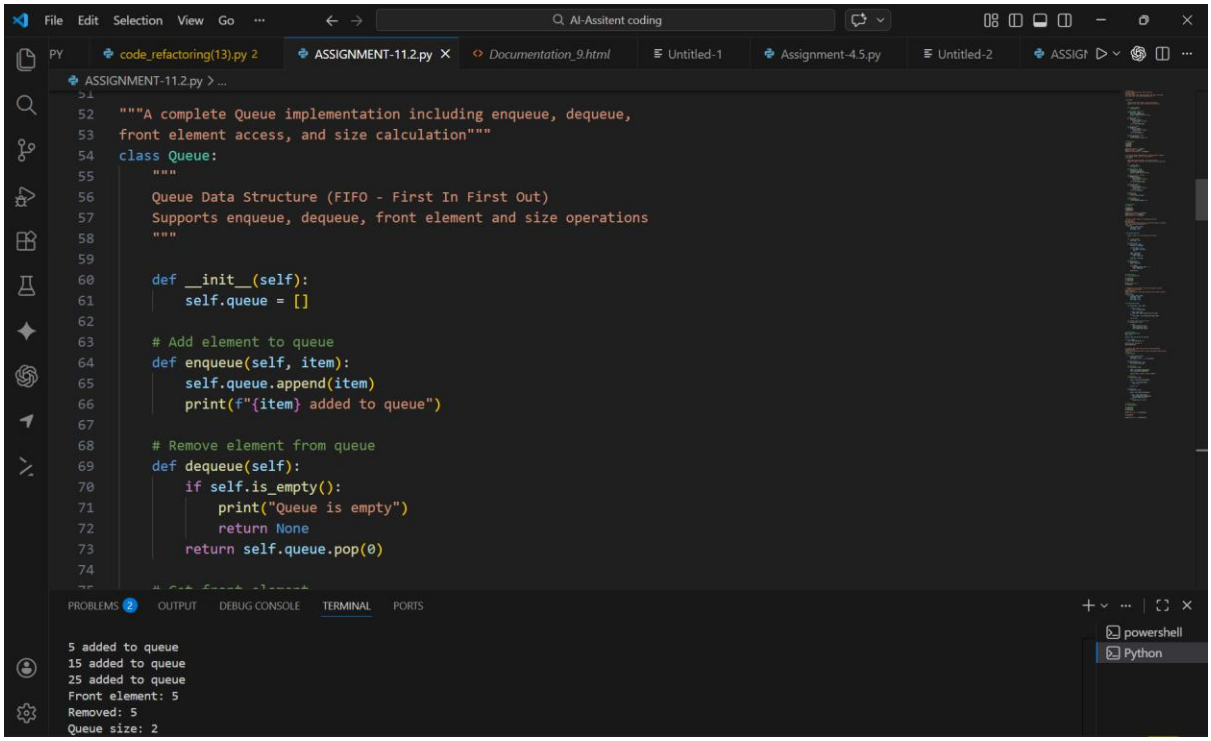
-Really well explained and understood concepts like:

-push,pop,peak,isempty with simplest examples(in whichpop is only a methos and rest are variables which are for description)

TASK-2:

PROMPT:”"""A complete Queue implementation including enqueue, dequeue, front element access, and size calculation"""

CODE:



OBSERVATION:

-This included(Queue):

-Enque-insertion

-Dequeue-Deletion

-Front-to find emptiness

-size-length

-is_empty

TASK-3:

PROMPT:

"""Build a singly linked list supporting insertion

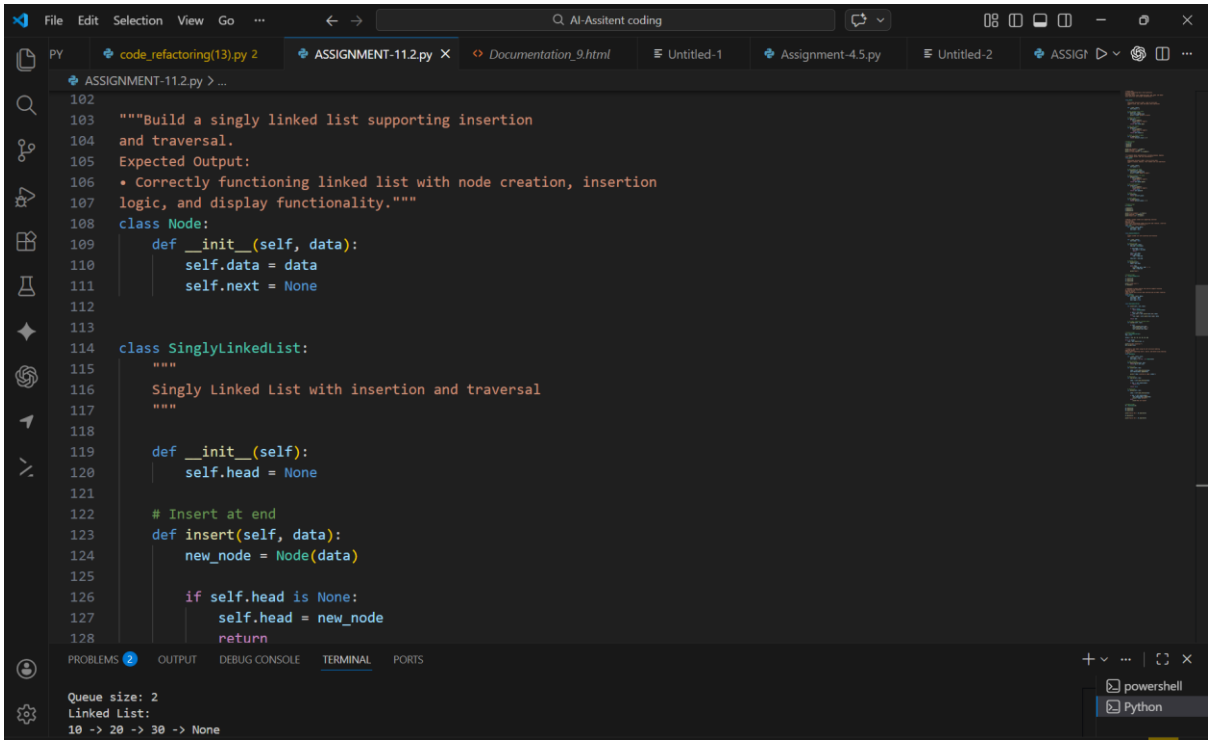
and traversal.

Expected Output:

• Correctly functioning linked list with node creation, insertion

logic, and display functionality."""

CODE:



OBSERVATION:

Build a simple single linked list

-insertion

-Traversal methods are used

TASK-4:

PROMPT:

"""Implement a Binary Search Tree with AI support focusing

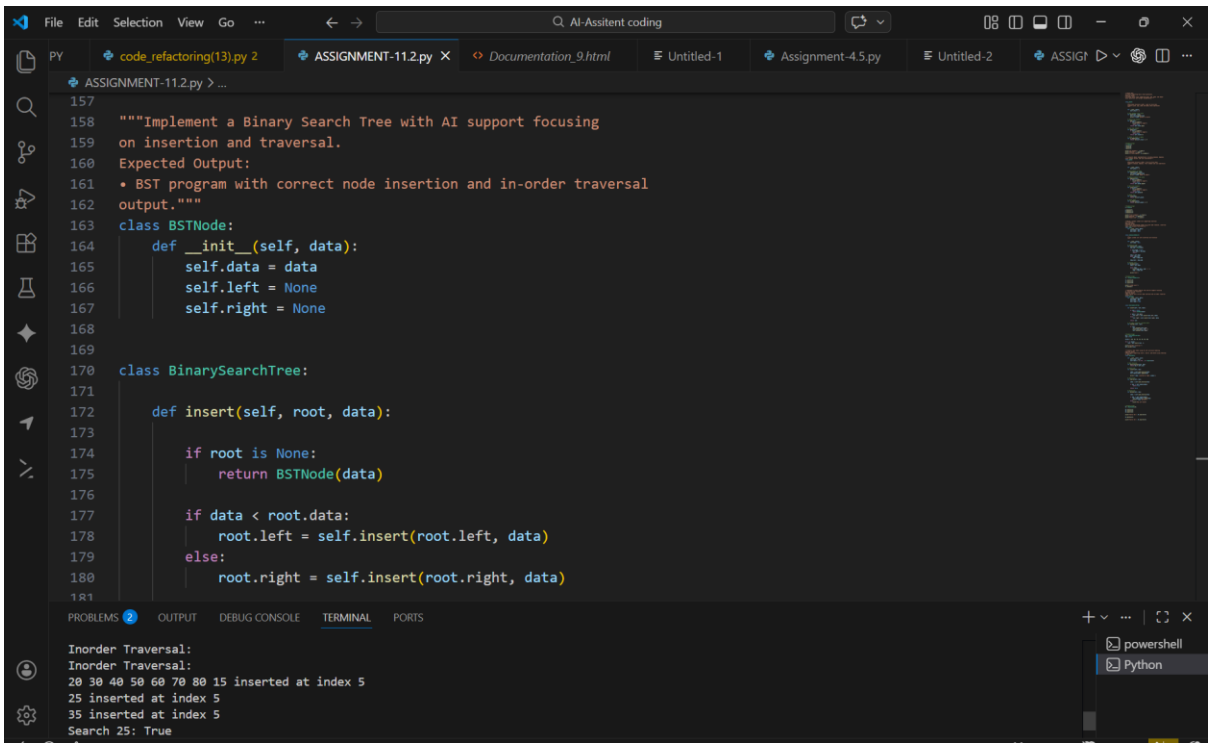
on insertion and traversal.

Expected Output:

• BST program with correct node insertion and in-order traversal

output."""

CODE:



OBSERVATION:

-Inoreder insertion is used

-BST used for quick insertion in Binary Tree

TASK-5:

PROMPT:

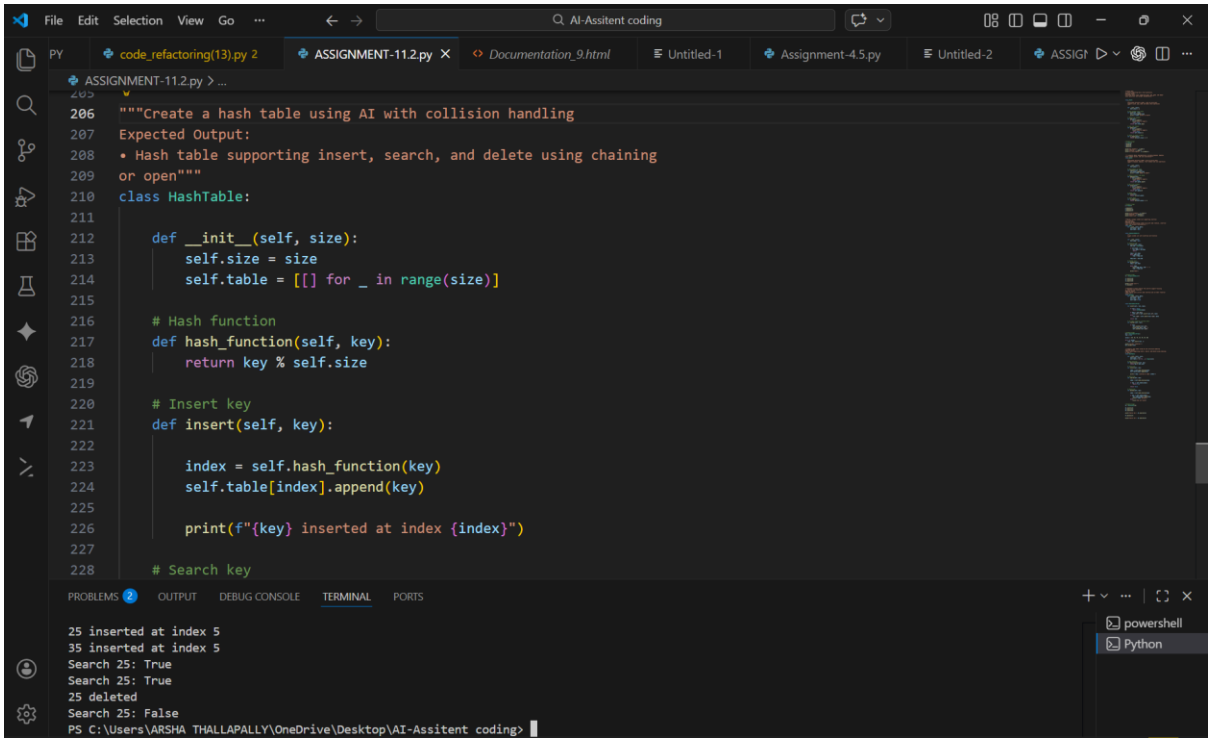
"""Create a hash table using AI with collision handling

Expected Output:

• Hash table supporting insert, search, and delete using chaining

or open"""

CODE:



OBSERVATION:

Hash Table created :

With functionalities like:Has_function-The main formula given

-Insertion of elements

-Search of numbers

-Deletion of Elements